

KOSTANAY, Kazakhstan

WMO: 289520

Lat: 53.2369N

Lon: 63.7408E

Elev: 170

StdP: 99.30

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.1	-28.9	-35.5	0.1	-32.0	-32.1	0.2	-28.6	12.5	-6.5	11.4	-7.5	2.2	20	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	32.9	18.5	31.0	18.3	29.1	17.9	20.8	28.4	19.9	27.6	19.1	26.6	4.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.4	23.4	17.2	12.6	22.6	16.2	11.8	22.1	60.6	28.4	57.8	27.7	54.9	26.5	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	10.0	8.9	DB	-35.4	35.7	3.6	1.9	-37.9	37.1	-40.0	38.2	-42.0	39.2	-44.6	40.6	(4)
(5)			WB	-35.5	22.9	3.5	1.5	-38.0	23.9	-40.1	24.8	-42.0	25.6	-44.6	26.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-16.1	-14.7	-6.3	5.9	13.9	19.1	20.3	18.9	12.6	4.9	-4.9	-12.8		
	DBStd	14.42	7.03	7.00	5.80	5.89	4.95	4.32	3.71	4.34	4.83	4.89	6.91	7.17		
	HDD10.0	3526	810	691	504	149	19	1	0	1	27	169	447	707		
	HDD18.3	5652	1068	924	762	374	152	42	22	47	181	416	697	966		
	CDD10.0	1153	0	0	0	24	141	273	321	278	104	12	0	0		
	CDD18.3	238	0	0	0	0	16	64	84	65	9	0	0	0		
	CDH23.3	2878	0	0	0	14	260	798	923	738	143	2	0	0		
	CDH26.7	1062	0	0	0	2	69	307	344	291	49	0	0	0		
(14) Wind	WSAvg	4.0	3.6	3.9	4.5	4.6	4.4	4.0	3.7	3.6	3.7	3.9	4.0	3.9		
(15) Precipitation	PrecAvg	334	17	14	16	24	31	38	57	35	26	29	24	23		
	PrecMax	476	40	36	38	67	91	105	151	101	76	109	51	59		
	PrecMin	220	0	1	0	0	0	4	6	0	2	0	0	1		
	PrecStd	68	10	9	11	16	20	27	37	24	20	21	13	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.8	1.7	10.9	25.1	31.1	34.2	35.8	35.0	32.1	22.1	10.2	2.2		
		MCWB	0.4	0.5	6.6	12.9	17.1	18.5	18.9	19.0	16.4	11.9	7.5	0.7		
	2%	DB	-1.3	0.2	4.8	21.9	28.8	32.2	32.3	32.2	27.0	18.3	7.0	0.7		
		MCWB	-1.9	-0.5	2.6	12.1	16.2	18.2	18.9	18.9	14.5	10.8	5.2	-0.1		
	5%	DB	-3.1	-1.6	2.8	18.9	26.3	30.2	30.1	30.0	24.0	15.5	5.1	-1.0		
		MCWB	-3.7	-2.3	1.4	10.8	15.1	17.9	18.7	18.7	14.0	9.2	3.5	-1.7		
	10%	DB	-5.2	-4.0	1.5	15.9	23.9	28.0	28.2	27.6	21.2	12.8	3.2	-2.9		
		MCWB	-5.7	-4.8	0.4	9.4	14.0	17.2	18.3	17.7	13.0	8.1	1.9	-3.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.3	0.7	6.6	14.5	19.2	21.1	22.2	21.6	18.2	13.0	7.8	1.1		
		MCDB	0.5	1.2	10.7	22.5	27.9	28.9	28.8	29.8	27.5	19.8	9.4	1.7		
	2%	WB	-1.9	-0.5	2.8	12.7	17.1	20.0	21.0	20.5	16.1	11.3	5.3	-0.3		
		MCDB	-1.5	0.1	4.3	20.1	26.3	28.9	27.7	29.1	24.2	17.2	6.6	0.4		
	5%	WB	-3.7	-2.3	1.5	11.3	15.9	19.1	20.1	19.5	14.8	9.8	3.6	-1.8		
		MCDB	-3.1	-1.7	2.5	17.7	24.2	27.4	26.7	27.4	22.3	14.7	4.8	-1.1		
	10%	WB	-5.7	-4.7	0.5	9.7	14.6	18.1	19.3	18.5	13.5	8.3	2.0	-3.5		
		MCDB	-5.2	-4.1	1.4	15.4	22.2	25.7	26.1	25.5	20.1	12.2	3.0	-2.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	10.0	9.4	11.5	12.9	12.6	11.6	12.3	12.3	10.0	7.6	8.6		
		MCDBR	8.2	8.6	8.4	16.5	16.7	15.8	14.8	15.8	16.8	15.2	8.4	7.8		
	5% WB	MCWBR	7.9	8.1	6.5	8.9	7.8	6.3	5.9	6.6	7.7	8.9	6.6	7.4		
		MCWBR	8.1	8.5	8.0	14.9	15.0	13.7	12.5	13.8	14.5	13.9	8.0	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.289	0.321	0.372	0.409	0.408	0.404	0.420	0.433	0.380	0.353	0.320	0.289		
	taud		2.080	2.026	1.978	2.134	2.226	2.313	2.265	2.202	2.302	2.283	2.194	2.067		
	Ebn at Noon		651	748	786	816	840	848	824	786	786	719	615	564		
	Edh at Noon		73	106	139	136	131	121	125	126	101	83	66	61		
(44) All-Sky Solar Radiation	RadAvg		1.15	2.28	3.79	4.72	5.74	6.22	5.86	4.84	3.48	1.86	0.93	0.74		
	RadStd		0.09	0.10	0.34	0.49	0.50	0.48	0.41	0.47	0.27	0.26	0.09	0.13		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days					
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)		
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)		

Nomenclature: See separate page